

Cosc 30: Discrete Mathematics

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Problem 1. Guessing game.

Imagine Alice and Bob are playing a guessing game. Alice chooses two distinct integers between 0 and 100, and the goal is for Bob to correctly guess which of the two numbers is bigger. Alice and Bob use the following protocol.

- Alice chooses two distinct integers between 0 and 100, writes the two numbers on separate pieces of paper, and puts the papers on the table facing down.
 - Bob chooses one paper to flip over, and then makes a guess on which number is bigger.
1. Assume that Alice chose the two integers uniformly at random. Come up with a strategy for Bob so that he can guess correctly with strictly more than 50% chance.
 - * 2. Now assume Alice gets to pick the two integers at will (in other words, her choice might not be uniformly at random). Come up with a strategy for Alice so that Bob, no matter which strategy he uses, will guess correctly with at most 51% chance.

Solution.

1. Let the number Bob sees be a , and the one he has to guess be b . Now, a strategy that results in $> 50\%$ chance of him guessing correctly is:

If $a < 50$, guess that the second number (b) is bigger. If $a > 50$ guess that the first number (a) is bigger. If $a = 50$, make a uniformly random guess about which of a and b is bigger.

Once Bob sees the first number (a), there are 100 possibilities for b . We have 3 cases:

(a) $a < 50$.

There are at least 51 numbers greater than a . Thus,

$$P[b > a] \geq \frac{51}{100}$$

That is he has at least a 51% chance of being correct.

(b) $\alpha > 50$.

There are at least 51 numbers smaller than α , so

$$P[b < \alpha] \geq \frac{51}{100}$$

That is he has at least a 51% chance of being correct.

(c) $\alpha = 50$

There are 50 numbers both greater and less than α . And Bob chooses greater or less with a 50% probability, hence, he has a 50% chance of being right.

However, because the numbers have a uniform probability of being chosen, we need to take a weighted average of the probabilities we have calculated.

$$\begin{aligned} P[\text{bob is right}] &\geq \frac{1}{101}P[\alpha = 50] + \frac{50}{101}P[\alpha > 50] + \frac{50}{101}P[\alpha < 50] \\ &\geq \frac{1}{101} \cdot \frac{1}{2} + \frac{50}{101} \cdot \frac{51}{100} + \frac{50}{101} \cdot \frac{51}{100} \\ &\geq \frac{103}{202} \approx .5099 \end{aligned}$$

Hence, Bob can guess correctly with strictly more than 50% chance.

2. Alice should pick an integer α , and then with uniform probability pick b to be either $\alpha + 1$ or $\alpha - 1$. Then, regardless of the strategy Bob uses, if the first number Bob sees is in the range $1 - 99$ there is only a 50% chance he guesses correctly, in the off-chance that he sees 0 or 100 he has a 100% chance of getting it right. But since Alice can choose any two integers of her liking, she should avoid the edge cases.

Problem 2. String compression reprise.

At the start of the term we showed that there is no lossless compression algorithm that guarantees to shorten every binary string. What if we are in the setting when the input files are sparse, that is, with a small number of 1s?

Let α be a constant strictly between 0 and 1/2. Let S_α denote the number of n -bit strings with at most an α fraction of the bits equal to 1. Using the binomial coefficient, we know that

$$S_\alpha = \sum_{0 \leq k \leq \alpha n} \binom{n}{k}.$$

Prove that asymptotically,

$$\binom{n}{\alpha n}$$

is roughly $2^{nH(\alpha) + \Theta(\ln n)}$, where

$$H(\alpha) := \alpha \ln_2 \frac{1}{\alpha} + (1 - \alpha) \ln_2 \frac{1}{1 - \alpha}$$

is the discrete entropy function. In other words, prove the following upper and lower bounds:

$$2^{nH(\alpha) + O(\ln n)} \geq \binom{n}{\alpha n} \geq 2^{nH(\alpha) - O(\ln n)}.$$

Because $S_\alpha \geq \binom{n}{\alpha n}$, this shows that no lossless compression algorithm can compress n -bit strings with at most an α fraction of 1s into fewer than $nH(\alpha) - O(\ln n)$ bits.

Solution.

We know from the formula for binomial coefficients:

$$\binom{n}{\alpha n} = \frac{n!}{(\alpha n)! \cdot (n - \alpha n)!}$$

Taking the natural logarithms for the expression, we get

$$\ln(n!) - \ln((\alpha n)!) - \ln((n - \alpha n)!)$$

which can be also written as:

$$\sum_{i=1}^n \ln(i) - \sum_{i=1}^{\alpha n} \ln(i) - \sum_{i=1}^{n-\alpha n} \ln(i)$$

The upper and lower bounds for each individual term is given by the integral approximation.

$\sum_{i=1}^n \ln(i)$ is an increasing function by definition as $\ln(y) > \ln(x)$ for $y > x$. Similarly, $\sum_{i=1}^{\alpha n} \ln(i)$ and $\sum_{i=1}^{n-\alpha n} \ln(i)$ are decreasing functions.

The approximations of the upper and lower for each summation is given by:

$$\begin{aligned} \int_1^n \ln(x) + \ln(1) &\leq \sum_{i=1}^n \ln(i) \leq \int_1^n \ln(x) + \ln(n) \\ -\int_1^{\alpha n} \ln(x) + (-\ln(\alpha n)) &\leq -\sum_{i=1}^{\alpha n} \ln(i) \leq -\int_1^{\alpha n} \ln(x) \\ -\int_1^{n-\alpha n} \ln(x) + (-\ln(n - \alpha n)) &\leq -\sum_{i=1}^{n-\alpha n} \ln(i) \leq -\int_1^{n-\alpha n} \ln(x) \log \end{aligned}$$

Calculating the integrals for each term:

$$\begin{aligned} \int_1^n \ln(x) &= [x \ln(x) - x]_1^n \\ &= n \ln(n) - n + 1 \end{aligned}$$

$$\begin{aligned} \int_1^{\alpha n} \ln(x) &= [x \ln(x) - x]_1^{\alpha n} \\ &= \alpha n \ln(\alpha n) - \alpha n + 1 \end{aligned}$$

$$\begin{aligned} \int_1^{n-\alpha n} \ln(x) &= [x \ln(x) - x]_1^{n-\alpha n} \\ &= (n-\alpha n) \ln(n-\alpha n) - (n-\alpha n) + 1 \\ &= (n-\alpha n) \ln(n) + (n-\alpha n) \ln(1-\alpha) - n + \alpha n + 1 \end{aligned}$$

Lower bound for the expression

$$\sum_{i=1}^n \ln(i) - \sum_{i=1}^{\alpha n} \ln(i) - \sum_{i=1}^{n-\alpha n} \ln(i)$$

is :

$$\begin{aligned} \text{LB} &= n \ln(n) - n - \alpha n \ln(\alpha n) + \alpha n - (n-\alpha n) \ln(1-\alpha) \\ &\quad - (n-\alpha n) \ln(n) + (n-\alpha n) \ln(1-\alpha) \end{aligned}$$

Simplifying this expression, we get:

$$\begin{aligned}
&= -\mathfrak{n}(1 - \alpha) \ln(1 - \alpha) - \alpha \mathfrak{n} \ln(\alpha) - 2 \ln(\mathfrak{n}) - \ln(\alpha) - \ln(1 - \alpha) \\
&= \alpha \mathfrak{n} \ln\left(\frac{1}{\alpha}\right) + \mathfrak{n}(1 - \alpha) \ln\left(\frac{1}{1 - \alpha}\right) - O(\ln \mathfrak{n}) \\
&= \mathfrak{n}H(\alpha) - O(\ln \mathfrak{n})
\end{aligned}$$

Similarly, after simplifying the expression for the upper-bound, we get:

$$\begin{aligned}
\text{UB} &= -\alpha \mathfrak{n} \ln(\alpha) - \mathfrak{n}(1 - \alpha) \ln(1 - \alpha) + \ln(\mathfrak{n}) \\
&= \alpha \mathfrak{n} \ln\left(\frac{1}{\alpha}\right) + \mathfrak{n}(1 - \alpha) \ln\left(\frac{1}{1 - \alpha}\right) + O(\ln \mathfrak{n}) \\
&= \mathfrak{n}H(\alpha) + O(\ln \mathfrak{n})
\end{aligned}$$

Taking the logarithmic expressions back into original form, we get:

$$2^{\mathfrak{n}H(\alpha) + O(\ln \mathfrak{n})} \geq \binom{\mathfrak{n}}{\alpha \mathfrak{n}} \geq 2^{\mathfrak{n}H(\alpha) - O(\ln \mathfrak{n})}$$